Radio October Range Test Plan

Goal: Using the existing teensy 4.1 boards we are testing their ground to ground range from an elevated position. The goal is to confirm that the 30Dbm Dorji (radio shields) that we are using + ideal antennas at the **435 band** will be able to reach at least 30km of range.

Bands: We are only testing at 435 since this years rocket is using only that band

Personnel: We are keeping this lean, and limited to project leads for smoothest operation + TingYi cause hes super experienced

Date: Oct 18th 2026

Team 1: Mock FC Team
Jeff (driver)
Nye
Martin

Team 2: Ground Station Radio Team
TingYi (Driver)
Yann
Max

Plan: Team1 will be at the mont St hilare, while Team2 tests reception from progressively farther ranges. We need to test commanding and receiving so it has to be a full ping pong.
Team2 should have the Yagi + Pad Whip

Testing method:
Use the same range test script, but we have less of a focus on the antennas, and care more about the absolute range

Team 1 Location:
Primary location will be to hike up the following trail, with a radio, with a simple whip antenna. This will be the mock FC.
https://www.alltrails.com/trailr/canada/quebec/boucle-du-pain-de-sucre-et-du-lac-hertel



Jeff can confirm from a previous range test that we have line of sight up to at least to lake chambly https://maps.app.goo.gl/9byi6b2GiG7Pyr9y6.



Team2 Locations:
(.5 Locations are optional if we see performance degradation, at a major point instead of hoping to the next major we will go to an in between point.

L0.5:
https://www.google.com/maps/place/1375+Chemin+des+Patriotes+Sud,+Otterburn+Park,+QC+J3H+4K7/@45.5358637,-73.2289431,19.5z/data=!4m6!3m5!1s0x4cc9aa5b9b2aa8e1:0xe7528b7b1298501a!8m2!3d45.5356769!4d-73.229061!16s%2Fg%2F11ghrk41ym?entry=ttu&g_ep=EgoyMDI2MTAxMy4wIKXMDSoASAFQAw%3D%3D
1375 Chemin des Patriotes Sud, Otterburn Park

Range: 5km

L1:* https://maps.app.goo.gl/zuPtVhgHcD9F4eY46
Parc Des Isles, Lac Chambly
Range: 11km

L2: https://maps.app.goo.gl/wqsJ6SdMJwht9acu5
Parc Dela Pointe, Along Richelieu River
Range: 16.24km

L2.5: Parc Gérard-Morin
Parc Gerard-Morin, Ile-Ste-Therese
Range 20.41km

L3: https://maps.app.goo.gl/xTEmeJ3aRXmjbUyw8
City park, Saint Jean sur le Richelieu
Range: 25.5km

L3.5: Dog Park at Centre-de-Plein-Air-Ronald-Beauregard
Dog Park, Saint-Jean-sur-Richelieu
Range: 29.73km

L4: https://maps.app.goo.gl/u9Lsz6L9uKxpHdZD8
Lake, Saint-Jacques-le-Mineur
Range: 34.81Km

L4.5:
https://www.google.com/maps/place/Quai+Ryan+de+Sabrevois/@45.196621,-73.2512573,17z/data=!4m6!3m5!1s0x4cc990ab17dabe3b:0x5c25226221c5d92e!8m2!3d45.1966172!4d-73.2486824!16s%2Fg%2F11c2jb5tx5?entry=ttu&g_ep=EgoyMDI2MTAxMy4wIKXMDSoASAFQAw%3D%3D
Quai Ryan de Sabrevois
Range: 39.7 km

L5: https://maps.app.goo.gl/e4tLgLTf1aX37oLM8
Dog park, Napierville
Range: 43Km

* At L1 we will try different radio configurations listed below and decide on one for the rest of the test. All configurations should work, idea is that we have data to show at 10km range the behaviors, or gains that we get by switching. If unsure we will likely stick with configuration #2.

(Note that at high spreading factors we need to enable the Low Data Rate Optimization (LDRO) in the sx126x, Radio lib handles the LDRO logic)

LoRa test configurations

Config	SF	BW (kHz)	Gain vs baseline	Airtime (42 B)	Purpose
0	8	250	0 dB	185 ms	Baseline
1	8	125	+3 dB	369 ms	Bandwidth effect
2	10	125	+8 dB	1,214 ms	Moderate range
3	10	62.5	+11 dB	2,953 ms	Extended range